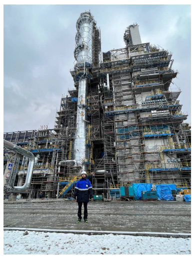
Denis Meleshkin

melesh.denis@gmail.com LinkedIn profile Portfolio
Padova, Italy

Curriculum vitae



Working Experience

2022 – 2023 **Jr. Design Engineer**. PJSC "Cryogenmash".
The main area of work was the calculation and modelling of cryogenic systems in Aspen HYSYS software, and rating of heat exchangers.

2021 – 2022 **Procurement technical engineer**. LLC STATMED.
The main responsibilities included: market analysis, interaction with partners, organization of international purchases of various volumes.

Internships

2022 **Production practice at the Amur Gas Processing Plant.**
During the internship the operation of the natural gas purification, separation and liquefaction line, as well as the production of liquid helium, was studied.

2021 **Technological practice at the NPO Nauka enterprise.**
During the internship a technological process for manufacturing a pressure valve part was developed.

Education

2023 – 2025 **M.Sc. Energy Engineering.**
University of Padua, Industrial Engineering department.
Thesis title: *Numerical investigation on the use of Twisted Tape turbulators for water single-phase forced convection*.
Final grade 100/110.

2018 – 2022 **B.Sc. Refrigeration, Cryogenic Equipment And Life Support Systems.**
Bauman Moscow State Technical University, Power Engineering department
Thesis title: *Helium cryogenic cooling system with a capacity of 2 kW per 100-150 K*.
Diploma with honour.

Publications

Journal Articles

- **D. Meleshkin**, S. Pochatkov, and I. Svetlov, "Vibrodiagnostics method for determining damage to composite materials," *Grand Mechanic*, no. 5, pp. 337–342, 2022. URL: https://www.elibrary.ru/item.asp?id=48445134.

Conference Proceedings

- D. Ershov, A. Mironov, **D. Meleshkin**, S. Pochatkov, and I. Svetlov, "Methods of numerical analysis and vibration diagnostics for assessing deformations in mechanical systems," in *Students' scientific spring (2022)*, Moscow, Russia, 2022, pp. 22–24. URL: https://www.elibrary.ru/item.asp?id=49480593.

Skills

Engineering	• Design and analysis of energy systems and devices, modelling and simulation; • Planning and caring out an experiment, processing the results; • Entropy and exergy analysis; • Life Cycle Assessment; • Developing of the design documentation.
Software	• Calculations: MATLAB, PTC Mathcad, MS Excel, Julia; • Modelling and simulations: Simulink and Simscape, ANSYS, OpenFOAM, Aspen HYSYS; • Design and documentation: Autodesk Inventor, Autodesk AutoCAD, Solid-Edge, CFturbo; • Reporting and publishing: LaTeX, MS Word, MS PowerPoint, MS Visio, GIT.
Language	• Russian: native speaker; • English: Strong reading, writing and speaking competencies, proved by IELTS; • Italian: basic proficiency with conversational ability.
Misc.	• Academic research, resource allocation, working in groups, management skills; • Leadership experience: during the studying for the Bachelor's degree, I was elected head of the study group, held this honorary position until graduation and was awarded a special payment for it.